Project Title: AIRiskWatch: An intelligence system for monitoring and analysis of AI incidents

Problem Statement: With the rapid growth of AI systems and GenAI applications, incidents involving harmful or unsafe AI behavior are increasing across different domains. These incidents include misinformation, deepfake misuse, biased decision-making, privacy leakage, hallucinations, unsafe recommendations, and many more. AI governance analysts, researchers and developers require insights into these incidents to better understand patterns and make more informed decisions and policies. These incidents also need to be organized using a standardized taxonomy so that organizations can describe AI risks consistently, track trends more accurately, and create better safety policies. However, information about such incidents are scattered across news articles, blogs, and other online sources. Even though some platforms may show visualizations of recent incidents and classify the incidents based on established taxonomies, they rely on static datasets and do not provide contextual explanation or semantic relationship analysis of these incidents and the AI systems that caused them.

Proposed Solution: AIRiskWatch is a monitoring and analysis system. The platform will be designed to automatically collect AI incident-related articles and reports from a selected sources and classify them according to one of the established taxonomies of AI harm. The extracted information- the AI system involved, the organization that developed it, the harmed parties, supporting reports of this incident- along with the taxonomy based classification information will be stored in a knowledge base that will represent the relationship between these entities. The platform will provide interactive dashboards and visualizations to help users explore AI incident trends based on affected sectors, stakeholder impacts, and relationships between risks and organizations. In addition to visualizations, the system will also generate contextual summaries and semantic search results to help users better understand the meaning and significance of the displayed information.

Project Description and Scope: The proposed platform will be a web-based application. It'll collect AI incident-related reports from selected news sources using a scraper. The scraped report will be processed and classified according to MIT AI Risk Taxonomy using a LLM. The structured information will be stored in a knowledge graph that captures relationships between the relevant entities- AI system, harmed parties/stakeholders, deployer of the system, etc. The platform will provide an interactive dashboard where users can explore incident trends, filter by risk category from the taxonomy, organization that deployed the AI, affected stakeholders, and time period. Users will also be able to search for incidents using natural language queries, with the system returning contextually relevant results and summaries powered by GraphRAG. An existing dataset from the AI Incident Database will be used to build the initial knowledge graph, and will grow automatically as new incidents are scraped from the selected web sources.